

Fine-Structure Constant: Unit Conventions Why  $\alpha = 1$  can  
be set Supplement to Document 011

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## Abstract

This document addresses aspects of the fine-structure constant not discussed in detail in Document 011. The focus is on providing a comprehensive justification for why and how  $\alpha = 1$  can be set (Heaviside-Lorentz convention), the physical consequences of different unit systems, and the historical and practical implications of redefining electromagnetic units.

**For T0-specific derivations** (characteristic energy  $E_0$ , geometric parameter  $\xi$ , T0 formula  $\alpha = \xi(E_0/1\text{MeV})^2$ ) see Document 011.

## .1 Introduction and Reference to Document 011

### Delimitation from Document 011

**Document 011** covers in detail:

- T0 derivation:  $\alpha = \xi(E_0/1\text{MeV})^2$
- Characteristic energy:  $E_0 = \sqrt{m_e \cdot m_\mu} = 7.348 \text{ MeV (bare)}$ , corrected  $E_0 = 7.348/\sqrt{K_{\text{frak}}} = 7.398 \text{ MeV}$
- Geometric parameter:  $\xi = \frac{4}{3} \times 10^{-4}$

- Alternative formulations: with  $\mu_0$ , with  $r_e/\lambda_C$ , etc.
- Historical context (Sommerfeld)
- Natural units and energy as fundamental field
- Detailed dimensional analysis of all formulations
- **This document (044)** focuses on:
  - **Why**  $\alpha = 1$  can be set (detailed justification)
  - **How** different unit conventions work
  - Consequences of redefining the Coulomb
  - Heaviside-Lorentz vs. Gauss vs. SI units
  - Practical aspects and historical development
  - Fine's inequality vs. fine-structure constant (name confusion)

## Why Two Documents?

**Document 011:** T0 theory and physical derivations

**Document 044:** Unit systems and conventions

Both complement each other, with minimal overlap.

## .2 Different Unit Conventions for $\alpha$

### Overview of Systems

The fine-structure constant can be expressed in different unit systems:

| System            | Formula                                                          | Value                   |
|-------------------|------------------------------------------------------------------|-------------------------|
| SI Standard       | $\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c}$                     | $\approx \frac{1}{137}$ |
| Heaviside-Lorentz | $\alpha = \frac{e^2}{4\pi}$ (with $\hbar = c = \epsilon_0 = 1$ ) | 1 or $\frac{1}{137}$    |
| Gauss (cgs)       | $\alpha = \frac{e^2}{\hbar c}$                                   | $\approx \frac{1}{137}$ |

**Table 1:** Unit systems for  $\alpha$

**Important:** The numerical value depends on the convention, the *physical* predictions do not.

## .3 Heaviside-Lorentz Units in Detail

### What are Heaviside-Lorentz Units?

The Heaviside-Lorentz system is a variant of natural units, specifically for electrodynamics:

$$\boxed{\hbar = c = \epsilon_0 = 1} \quad (1)$$

#### Consequences:

- Electromagnetic equations become more symmetric
- The factor  $4\pi$  disappears from many formulas
- Elementary charge is redefined

**Why  $\varepsilon_0 = 1$ ?**

In SI units,  $4\pi$  appears in many electromagnetic formulas:

$$\vec{E} = \frac{1}{4\pi\varepsilon_0} \frac{q}{r^2} \hat{r} \quad (\text{Coulomb's law}) \quad (2)$$

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \quad (\text{Maxwell's equation}) \quad (3)$$

With  $\varepsilon_0 = 1$ , these become:

$$\vec{E} = \frac{q}{4\pi r^2} \hat{r} \quad (4)$$

$$\nabla \cdot \vec{E} = \rho \quad (5)$$

The factor  $4\pi$  remains only in Coulomb's law (sphere surface  $4\pi r^2$ ); Maxwell's equations become free of  $4\pi$ .

**Fine-Structure Constant in Heaviside-Lorentz**

**Starting point (SI):**

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} \quad (6)$$

**With  $\hbar = c = \varepsilon_0 = 1$ :**

$$\alpha = \frac{e^2}{4\pi \cdot 1 \cdot 1 \cdot 1} = \frac{e^2}{4\pi} \quad (7)$$

**Now the crucial question:** What value does  $e$  have in this system?

**.4 Two Variants of Heaviside-Lorentz**

**Variant A: Normalize  $e$  so that  $\alpha = 1$**

**Approach:** We define the unit of charge such that  $\alpha = 1$ .

Since  $\alpha = e^2/(4\pi)$  in HL units:

$$e^2 = 4\pi \Rightarrow e = \sqrt{4\pi} \approx 3.545 \quad (8)$$

**Physical meaning:**

- Elementary charge becomes a *dimensionless unit*
- Electromagnetic coupling is "normalized"
- Charge is measured in units of  $\sqrt{\hbar c}$

**What changes?**

The elementary charge gets a new numerical value:

$$e_{\text{HL}} = \sqrt{4\pi\varepsilon_0\hbar c} \quad (\text{expressed in SI units}) \quad (9)$$

Numerically:

$$e_{\text{HL}} = \sqrt{4\pi \times 8.854 \times 10^{-12} \times 1.055 \times 10^{-34} \times 3 \times 10^8} \quad (10)$$

$$\approx 1.877 \times 10^{-18} \text{ C} \quad (11)$$

This is  $\sqrt{137.036} \times e_{\text{SI}}$  – the charge for which  $\alpha = 1$ . In the HL charge unit  $\sqrt{\varepsilon_0\hbar c} \approx 5.29 \times 10^{-19} \text{ C}$  it has the numerical value  $\sqrt{4\pi}$ .

**Variant B: Keep  $e$ ,  $\alpha \approx 1/137$** **Approach:** The elementary charge retains its "natural" value.

In this case:

$$\alpha = \frac{e^2}{4\pi} \approx \frac{1}{137} \quad (12)$$

because  $e$  in these units has the value  $\sqrt{4\pi/137} \approx 0.303$ .**Physical meaning:**

- Charge retains physical meaning
- $\alpha$  remains  $\approx 1/137$
- Only the mathematical form simplifies

**Which variant is used?****In practice:**

- **T0 theory:** Sets all constants = 1 ( $c = \hbar = \alpha = G = 1$ )
- **Theoretical high-energy physics:** Often  $\hbar = c = 1$ , sometimes also  $\alpha = 1$
- **Numerical calculations:** Often  $\hbar = c = 1$ , but  $\alpha \approx 1/137$
- **Experimental physics:** Almost always SI units (all constants have numerical values)
- **T0 convention:**
  - In T0 calculations:  $c = \hbar = \alpha = G = 1$  (maximum simplification)
  - Only parameter:  $\xi = \frac{4}{3} \times 10^{-4}$  (integer structural choices are motivated settings; SI anchors such as  $m_e$  serve for conversion)
  - When comparing with experiments: SI values ( $c = 3 \times 10^8$  m/s,  $\alpha \approx 1/137$ , etc.)
  - Both describe the same physics.

**.5 Reconstruction of SI Values from T0****The Central Principle****Basic idea:** Although T0 sets all constants to 1, the SI values can be reconstructed.**T0 Reconstruction****In T0 calculations:**

- All constants = 1:  $c = \hbar = \alpha = G = 1$
- Only parameter:  $\xi = \frac{4}{3} \times 10^{-4}$  (plus structural choices as settings)
- Formulas maximally simplified

**Reconstruction of SI values:**

- Fine-structure constant:  $\alpha_{\text{SI}} = \xi(E_0/1\text{MeV})^2 \approx 1/137$
- Gravitational constant:  $G_{\text{SI}} = \frac{\xi^2}{4m_e} \times \text{factors}$
- Further constants: reconstructable from  $\xi$  and energy scales (see Doc. 011 and 012)

### Example: Fine-Structure Constant

In T0 units:

$$\alpha = 1 \quad (13)$$

**Reconstruction of SI value:**

$$\alpha_{\text{SI}} = \xi \cdot \left( \frac{E_0}{1 \text{ MeV}} \right)^2 \quad (14)$$

With  $\xi = \frac{4}{3} \times 10^{-4}$  and  $E_0 = 7.398 \text{ MeV}$ :

$$\alpha_{\text{SI}} = 1.3333 \times 10^{-4} \times (7.398)^2 \quad (15)$$

$$= 1.3333 \times 10^{-4} \times 54.73 \quad (16)$$

$$= 7.297 \times 10^{-3} \quad (17)$$

$$= \frac{1}{137.04} \quad (18)$$

Experimental:  $\alpha_{\text{exp}} = \frac{1}{137.036}$

Deviation from the experimental value: 0.0005 % (unrounded  $1/(1.3333 \times 10^{-4} \times 54.730) = 1/137.035$ ); even the rounded value 137.04 deviates by only 0.003 %.

### Example: Gravitational Constant

In T0 units:

$$G = 1 \quad (19)$$

**Reconstruction of SI value:**

$$G_{\text{SI}} = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}} \quad (20)$$

where:

- $C_{\text{conv}}$  = Conversion factors (eV  $\rightarrow$  J, etc.)
- $K_{\text{frak}}$  = fractal correction

A separate dimension conversion  $C_{\text{dim}} = 1/E_{\text{char}} = 3.472 \times 10^{-2}$  does not appear: it is cancelled again by the factor  $E_{\text{char}} = 28.8$  in the step chain of Doc. 012; inserted into the formula on its own it would give  $2.35 \times 10^{-12}$  instead of  $6.674 \times 10^{-11}$ .

Detailed derivation see Document 012 (Gravitation).

### Why does this work?

**Key:**  $\xi$  is dimensionless and is taken as universal in the T0 framework.

1. In T0:  $\xi$  determines all coupling strengths
2. In SI:  $\xi$  together with characteristic energies ( $E_0$ , masses) reconstructs all constants
3. Physical predictions: identical in both systems
4. Only the mathematical representation differs

| Constant | T0                           | SI                                                            | Reconstruction                       |
|----------|------------------------------|---------------------------------------------------------------|--------------------------------------|
| $c$      | 1                            | $3 \times 10^8 \text{ m/s}$                                   | Convention                           |
| $\hbar$  | 1                            | $1.055 \times 10^{-34} \text{ J}\cdot\text{s}$                | Convention                           |
| $\alpha$ | 1                            | $\approx 1/137$                                               | $\xi(E_0/1\text{MeV})^2$             |
| $G$      | 1                            | $6.67 \times 10^{-11} \text{ m}^3/(\text{kg}\cdot\text{s}^2)$ | $\xi^2/(4m_e) \times \text{factors}$ |
| $\xi$    | $\frac{4}{3} \times 10^{-4}$ | $\frac{4}{3} \times 10^{-4}$                                  | same                                 |

Table 2: T0 vs. SI - Reconstruction of constants

## Comparison Table

### Important Conclusion

- **T0 is not new physics**, but a reparametrization
- Instead of many constants ( $c, \hbar, \alpha, G, \dots$ ) only **one parameter**  $\xi$
- All SI values reconstructable from  $\xi$  and energy scales
- Advantage: Formulas simpler, physical relationships clearer
- Disadvantage: Conversion to SI needed for experiments

## .6 Why can $\alpha = 1$ be set?

### Fundamental Insight

#### Core Statement

The fine-structure constant  $\alpha$  is a **dimensionless number**. Its numerical value is **convention-dependent**, not fundamental.  
One can set  $\alpha = 1$  by redefining the **unit of charge** accordingly.

### Step-by-Step Justification

#### Step 1: What is a convention?

SI units are historically evolved definitions:

- 1 meter = originally 1/10,000,000 of the Earth's meridian
- 1 second = originally 1/86,400 of a solar day
- 1 Coulomb = defined via Ampere and force between currents

None of these is "fundamental".

#### Step 2: $\alpha$ in SI

In SI units:

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137} \quad (21)$$

The value  $1/137$  follows from:

- How we defined the Coulomb (historically)
- How we defined  $\epsilon_0$  (via  $\mu_0$  and  $c$ )

#### Step 3: Redefinition

We can say: "From now on, the elementary charge is no longer  $1.602 \times 10^{-19} \text{ C}$ , but  $e = \sqrt{4\pi\epsilon_0\hbar c}$ ."



Then automatically:

$$\alpha = \frac{(\sqrt{4\pi\epsilon_0\hbar c})^2}{4\pi\epsilon_0\hbar c} = 1 \quad (22)$$

#### Step 4: Physical Consequences

- **No physical prediction changes.**
- Only the *numbers* in formulas change
- All ratios remain the same
- All experiments yield the same results

### Analogy: Temperature Scales

**Celsius:** Water freezes at 0°C

**Fahrenheit:** Water freezes at 32°F

**Kelvin:** Water freezes at 273.15 K

None of these scales is more "correct" than the others; they are conventions.

Similarly,  $\alpha = 1/137$  (SI) vs.  $\alpha = 1$  (HL) is a choice of convention.

## .7 Consequences of Redefining the Coulomb

### What does it mean to redefine elementary charge?

If  $e$  is redefined such that  $\alpha = 1$ :

**Old definition (SI):**

$$e = 1.602 \times 10^{-19} \text{ C} \quad (23)$$

**New definition (HL with  $\alpha = 1$ ):**

$$e = \sqrt{4\pi} \approx 3.545 \quad (\text{dimensionless in natural units}) \quad (24)$$

or expressed in SI units:

$$e_{\text{new}} = \sqrt{4\pi\epsilon_0\hbar c} \approx 1.877 \times 10^{-18} \text{ C} = \sqrt{137.036} e \quad (25)$$

The charge unit is rescaled by the same factor:  $1 \text{ C}_{\text{new}} = (e_{\text{new}}/e) \cdot 1 \text{ C} = \sqrt{137} \times 1 \text{ C}_{\text{old}}$ . Numerical values transform inversely to the units: a unit larger by a factor  $f$  gives a numerical value smaller by  $f$ . (Here  $f$  is the scaling factor between the old and the new unit, not a frequency and not the base winding number  $f = 1/\xi$ .)

### Effects on Electromagnetic Quantities

#### Electric Current (Ampere)

Since  $1 \text{ A} = 1 \text{ C/s}$ :

$$1 \text{ A}_{\text{new}} = \frac{1 \text{ C}_{\text{new}}}{1 \text{ s}} = \sqrt{137} \times 1 \text{ A}_{\text{old}} \quad (26)$$

#### Electric Voltage (Volt)

$1 \text{ V} = 1 \text{ J/C}$ :

$$1 \text{ V}_{\text{new}} = \frac{1 \text{ J}}{1 \text{ C}_{\text{new}}} = \frac{1}{\sqrt{137}} \times 1 \text{ V}_{\text{old}} \quad (27)$$

**Capacitance (Farad)**

$$1 \text{ F}_{\text{new}} = \frac{1 \text{ C}_{\text{new}}}{1 \text{ V}_{\text{new}}} = 137 \times 1 \text{ F}_{\text{old}} \quad (28)$$

**Are these changes "real"?**

**No.** They are conversion factors, like Celsius  $\rightarrow$  Fahrenheit.

**All physical ratios remain identical:**

- Capacitance of a capacitor / distance: same
- Force between charges / distance<sup>2</sup>: same
- All experiments: same results

Only the *numerical values* we calculate with change.

**.8 Practical Impacts on Everyday Calculations****Motivation**

**Question:** If we set  $\alpha = 1$ , what does that mean for ordinary electrical calculations with volts, amperes, resistance, capacitance?

**Answer:** All formulas change, but the *physical results* remain identical.

**Example 1: Ohm's Law****In SI Units (Standard)**

$$U = R \cdot I \quad (29)$$

**Numerical example:**

- Resistance:  $R = 100 \, \Omega$
- Current:  $I = 2 \text{ A}$
- Voltage:  $U = 100 \times 2 = 200 \text{ V}$

**In Heaviside-Lorentz with  $\alpha = 1$** 

The formula remains  $U = R \cdot I$ , but the *numerical values* change.

**Unit conversion:**

$$1 \text{ A}_{\text{new}} = \sqrt{137} \times 1 \text{ A}_{\text{old}} \approx 11.7 \text{ A}_{\text{old}} \quad (30)$$

$$1 \text{ V}_{\text{new}} = \frac{1}{\sqrt{137}} \times 1 \text{ V}_{\text{old}} \approx 0.085 \text{ V}_{\text{old}} \quad (31)$$

$$1 \, \Omega_{\text{new}} = \frac{1}{137} \times 1 \, \Omega_{\text{old}} \quad (32)$$

**Same circuit in new units:**

$$R_{\text{new}} = 100 \times 137 = 13\,700 \, \Omega_{\text{new}} \quad (33)$$

$$I_{\text{new}} = \frac{2}{\sqrt{137}} \approx 0.171 \text{ A}_{\text{new}} \quad (34)$$

$$U_{\text{new}} = 13\,700 \times 0.171 = 2341 \text{ V}_{\text{new}} \quad (35)$$

**Conversion back to SI:**

$$U = 2341 \times 0.0854 \text{ V}_{\text{old}} = 200 \text{ V} \quad \checkmark \quad (36)$$

The result is identical.

**Example 2: Power of a Light Bulb****In SI Units**

$$P = U \cdot I = \frac{U^2}{R} \quad (37)$$

**Light bulb:** 60 W at 230 V

$$R = \frac{U^2}{P} = \frac{(230)^2}{60} = 882 \, \Omega \quad (38)$$

$$I = \frac{P}{U} = \frac{60}{230} = 0.26 \text{ A} \quad (39)$$

**In Heaviside-Lorentz with  $\alpha = 1$** 

**Power:** 1 W<sub>new</sub> = 1 W<sub>old</sub> (energy/time does not change in HL)

$$U_{\text{new}} = 230 \times \sqrt{137} = 2692 \text{ V}_{\text{new}} \quad (40)$$

$$R_{\text{new}} = 882 \times 137 = 1.21 \times 10^5 \, \Omega_{\text{new}} \quad (41)$$

$$I_{\text{new}} = \frac{P}{U_{\text{new}}} = \frac{60}{2692} = 0.0223 \text{ A}_{\text{new}} \quad (42)$$

**Verification:**

$$P = U_{\text{new}} \cdot I_{\text{new}} = 2692 \times 0.0223 = 60 \text{ W} \quad \checkmark \quad (43)$$

**Example 3: Charging a Capacitor****In SI Units**

$$Q = C \cdot U \quad (44)$$

**Capacitor:**  $C = 100 \, \mu\text{F}$  at  $U = 12 \text{ V}$

$$Q = 100 \times 10^{-6} \times 12 = 1.2 \times 10^{-3} \text{ C} \quad (45)$$

**Stored energy:**

$$E = \frac{1}{2} C U^2 = \frac{1}{2} \times 100 \times 10^{-6} \times 144 = 7.2 \times 10^{-3} \text{ J} \quad (46)$$

**In Heaviside-Lorentz with  $\alpha = 1$** **Conversion:**

$$1 \text{ F}_{\text{new}} = 137 \times 1 \text{ F}_{\text{old}} \quad (47)$$

$$1 \text{ C}_{\text{new}} = \sqrt{137} \times 1 \text{ C}_{\text{old}} \quad (48)$$

$$C_{\text{new}} = \frac{100 \times 10^{-6}}{137} = 7.30 \times 10^{-7} \text{ F}_{\text{new}} \quad (49)$$

$$U_{\text{new}} = 12 \times \sqrt{137} = 140.5 \text{ V}_{\text{new}} \quad (50)$$

$$Q_{\text{new}} = 7.30 \times 10^{-7} \times 140.5 = 1.025 \times 10^{-4} \text{ C}_{\text{new}} \quad (51)$$

**Conversion back:**

$$Q = 1.025 \times 10^{-4} \times \sqrt{137} = 1.2 \times 10^{-3} \text{ C}_{\text{old}} \quad \checkmark \quad (52)$$

**Energy:**

$$E_{\text{new}} = \frac{1}{2} \times 7.30 \times 10^{-7} \times (140.5)^2 = 7.2 \times 10^{-3} \text{ J} \quad \checkmark \quad (53)$$

The energy is the same in all systems.

## Example 4: RC Time Constant

**In SI Units**

$$\tau = R \cdot C \quad (54)$$

**Circuit:**  $R = 1 \text{ k}\Omega$ ,  $C = 10 \text{ }\mu\text{F}$

$$\tau = 1000 \times 10 \times 10^{-6} = 0.01 \text{ s} = 10 \text{ ms} \quad (55)$$

**In Heaviside-Lorentz with  $\alpha = 1$**

$$R_{\text{new}} = 1000 \times 137 = 1.37 \times 10^5 \text{ }\Omega_{\text{new}} \quad (56)$$

$$C_{\text{new}} = \frac{10 \times 10^{-6}}{137} = 7.30 \times 10^{-8} \text{ F}_{\text{new}} \quad (57)$$

$$\tau_{\text{new}} = 1.37 \times 10^5 \times 7.30 \times 10^{-8} = 0.01 \text{ s} = 10 \text{ ms} \quad \checkmark \quad (58)$$

**Time remains the same:** physical timescales do not change.

## Summary of Practical Calculations

| Quantity        | SI       | HL Factor      | HL ( $\alpha = 1$ )    |
|-----------------|----------|----------------|------------------------|
| Charge (Q)      | C        | $\sqrt{137}$   | $\sqrt{137} \text{ C}$ |
| Current (I)     | A        | $\sqrt{137}$   | $\sqrt{137} \text{ A}$ |
| Voltage (U)     | V        | $1/\sqrt{137}$ | $V/\sqrt{137}$         |
| Resistance (R)  | $\Omega$ | $1/137$        | $\Omega/137$           |
| Capacitance (C) | F        | 137            | 137 F                  |
| Power (P)       | W        | 1              | W (unchanged)          |
| Energy (E)      | J        | 1              | J (unchanged)          |
| Time ( $\tau$ ) | s        | 1              | s (unchanged)          |

**Table 3:** Conversion factors SI  $\rightarrow$  HL with  $\alpha = 1$  (factors for the units; numerical values are divided by the factor)

## Important Insights

### Core Statement

#### What changes:

- Numerical values for charge, current, voltage, resistance, capacitance

#### What does NOT change:

- Energy
- Power
- Time
- All physical ratios
- All experimental results

**Conclusion:** It is a conversion, comparable to meters  $\leftrightarrow$  feet.

## Why does nobody use $\alpha = 1$ in practice?

### Reasons:

1. **Measuring devices:** All voltmeters, ammeters, etc. are calibrated in SI
2. **Standards:** Worldwide accepted SI definitions
3. **Intuition:** Engineers know typical values in SI
  - Household: 230 V, not 2692 V<sub>new</sub>
  - USB: 5 V, not 58.5 V<sub>new</sub>
4. **Conversion is laborious:**  $\sqrt{137}$  factors everywhere
5. **No advantage for practitioners:** Simplification only visible in theoretical formulas  
**But:** In theoretical calculations (QED, Feynman diagrams),  $\alpha = 1$  can be helpful.

## .9 Practical Aspects of Different Systems

### Advantages and Disadvantages: SI Units

#### Advantages:

- Worldwide standardized
- Directly usable for experiments
- All measuring devices calibrated in SI
- Clear separation of length/time/mass/charge

#### Disadvantages:

- Many constants in formulas ( $4\pi\epsilon_0$ ,  $\hbar$ ,  $c$ )
- Physical relationships obscured
- Dimensions unwieldy

### Advantages and Disadvantages: Heaviside-Lorentz with $\alpha = 1$

#### Advantages:

- Maximally simplified formulas

- Electromagnetic symmetry visible
- Theoretical calculations simpler
- QED Feynman diagrams easier to read

**Disadvantages:**

- No direct connection to experiments
- Conversion to SI laborious
- Unfamiliar for practitioners
- Physical "size" of  $e$  unclear

**Advantages and Disadvantages: Natural Units with  $\alpha \approx 1/137$**

**Advantages:**

- Simplified formulas ( $\hbar = c = 1$ )
- $\alpha$  retains physical meaning
- Good compromise theory/practice
- Numerically:  $\alpha \ll 1 \rightarrow$  perturbation theory

**Disadvantages:**

- Still need conversion to SI
  - Factor  $4\pi$  remains in some formulas
- This is the usual convention in present-day particle physics.

## .10 Historical Development

### Gauss Units (cgs)

**19th century:** Gauss system (centimeter-gram-second)

$$\alpha = \frac{e^2}{\hbar c} \quad (59)$$

No  $4\pi\epsilon_0$ , because  $4\pi\epsilon_0 = 1$  by definition in cgs.

### SI Units (MKSA)

**20th century:** SI system (meter-kilogram-second-ampere)

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \quad (60)$$

The  $4\pi\epsilon_0$  appears because the SI ampere is defined via force.

### Heaviside-Lorentz

**Theoretical physics:** Heaviside-Lorentz simplifies Maxwell's equations

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \quad \nabla \times \vec{B} = \vec{j} + \frac{\partial \vec{E}}{\partial t} \quad (61)$$

The equations are symmetric (in SI,  $\mu_0$  and  $\epsilon_0$  appear asymmetrically)

## Natural Units

**Modern high-energy physics:**  $\hbar = c = 1$ , but different conventions for  $\alpha$

## .11 Fine's Inequality vs. Fine-Structure Constant

### Frequent Confusion

**Note:** *Fine's inequality* and the *fine-structure constant* are different concepts.

### Fine's Inequality

**What it is:**

- A form of Bell's inequality
- Test for local hidden variables
- Quantum entanglement vs. classical correlations

**Mathematically:**

$$|C(\alpha, \beta) - C(\alpha, \beta')| + |C(\alpha', \beta) + C(\alpha', \beta')| \leq 2 \quad (62)$$

where  $C$  are correlation functions.

**Physically:** Shows non-locality of quantum mechanics

### Fine-Structure Constant

**What it is:**

- Fundamental physical constant
- Strength of the electromagnetic interaction
- Dimensionless,  $\alpha \approx 1/137$

**Mathematically:**

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \quad (63)$$

**Physically:** Determines EM coupling strength

### No Conceptual Connection

The similarity in name is coincidental; the two concepts are not related.

## .12 Summary

### Core Statements

1.  $\alpha$  is dimensionless  $\rightarrow$  numerical value is convention-dependent
2. One **can** set  $\alpha = 1$  by redefining the unit of charge
3. **T0 theory:** Sets all constants = 1:  $c = \hbar = \alpha = G = 1$
4. Only parameter in T0:  $\xi = \frac{4}{3} \times 10^{-4}$  (integer structural choices are settings, SI anchors serve for conversion)
5. **No** physical prediction changes

6. Only numerical values in formulas differ
7. When comparing with experiments: SI values ( $\alpha \approx 1/137$ ,  $c = 3 \times 10^8$  m/s, etc.)

### For Further Details See Document 011

- T0 derivation of  $\alpha$
- Characteristic energy  $E_0$
- Geometric parameter  $\xi$
- Experimental verification
- Detailed dimensional analysis
- Historical context (Sommerfeld)

## 0.1 Conversion Table: SI $\leftrightarrow$ Heaviside-Lorentz

| Quantity                | SI                            | HL ( $\alpha = 1$ )             |
|-------------------------|-------------------------------|---------------------------------|
| Elementary charge       | $e = 1.602 \times 10^{-19}$ C | $e = \sqrt{4\pi} \approx 3.545$ |
| Fine-structure constant | $\alpha \approx 1/137$        | $\alpha = 1$                    |
| $\epsilon_0$            | $8.854 \times 10^{-12}$ F/m   | 1                               |
| $\hbar$                 | $1.055 \times 10^{-34}$ J·s   | 1                               |
| $c$                     | $3 \times 10^8$ m/s           | 1                               |

**Table 4:** Conversion table SI to HL

## 0.2 Sample Calculation: Coulomb's Law

### In SI Units

$$F = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} \quad (64)$$

Numerically for  $r = 1 \text{ \AA} = 10^{-10} \text{ m}$ :

$$F = \frac{1}{4\pi \times 8.854 \times 10^{-12}} \frac{(1.602 \times 10^{-19})^2}{(10^{-10})^2} \quad (65)$$

$$\approx 2.3 \times 10^{-8} \text{ N} \quad (66)$$

### In HL Units ( $\hbar = c = \epsilon_0 = 1$ )

$$F = \frac{e^2}{4\pi r^2} = \frac{\alpha}{r^2} \quad (67)$$

With  $r$  in natural units:  $r = 1 \text{ \AA} = 5.07 \times 10^{-4} \text{ eV}^{-1}$  (with  $\hbar c = 1.973 \times 10^{-7} \text{ eV m}$ )

$$F = \frac{\alpha}{(5.07 \times 10^{-4})^2} = \frac{3.89 \times 10^6}{137.036} \approx 2.84 \times 10^4 \text{ eV}^2 \quad (68)$$

Conversion to SI:  $1 \text{ eV}^2 = (1 \text{ eV})^2 / (\hbar c) \approx 8.12 \times 10^{-13} \text{ N}$



$$F \approx 2.3 \times 10^{-8} \text{ N} \quad (69)$$

**The result is identical;** only the intermediate steps look different.

The SI result  $2.3 \times 10^{-8} \text{ N}$  follows only with the numerical value  $\alpha = 1/137.036$  in the numerator; with  $\alpha = 1$  one would get  $3.2 \times 10^{-6} \text{ N}$ . The force depends on the physical value of  $\alpha$ , not on the choice of charge unit.